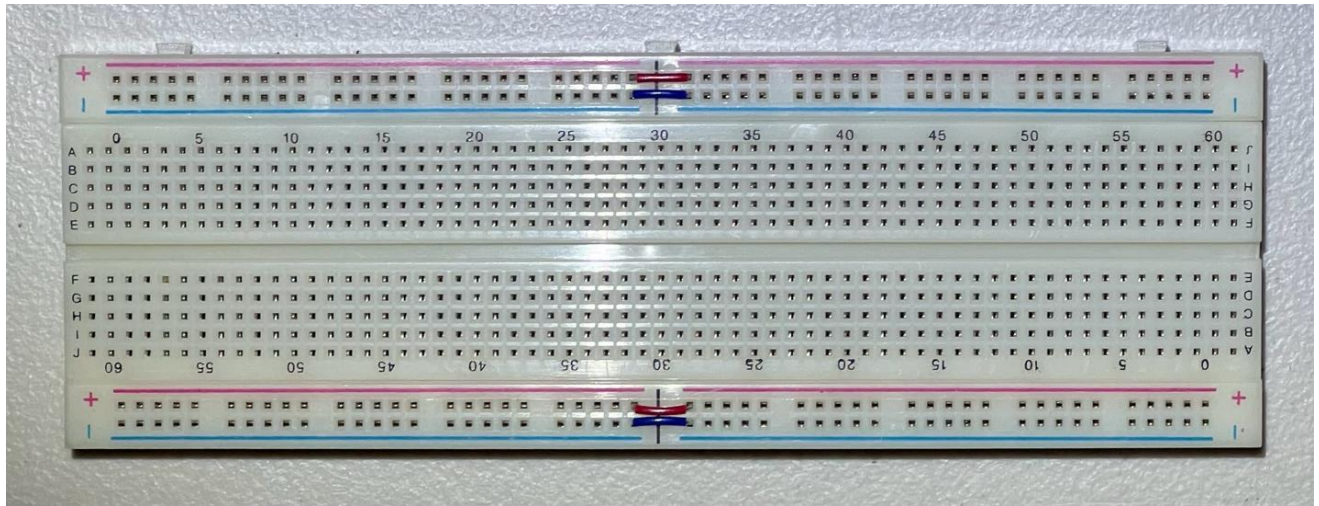


CS 350 Module One Lab Guide

As you work through the projects in this class, you will use a solderless breadboard to assemble some electronic components for your Raspberry Pi. Solderless breadboards allow you to assemble the components temporarily to prototype certain configurations without committing to building a permanent circuit board.

Below is a picture of the solderless breadboard included in this class's component kit and an explanation of how it works.



You will see that the solderless breadboard is divided into four distinct sets of columns. The two columns on either side labeled with + and – are called bus bars. They are connected vertically.

With the connector you are using for the Raspberry Pi, the bus bars on the left will be 3.3 volts and ground, and the bus bars on the right will be 5 volts and ground. This is important, as different components will require different voltages.

Each of the two sets of columns in the middle of the solderless breadboard are oriented differently, as they are tied together horizontally. As you can see, each column of pins is labeled with letters A–J. In this case, columns A, B, C, D, and E are tied together, as are columns F, G, H, I, and J. Because of this, anything plugged into an adjacent hole on the same row makes an electrical connection.

In Milestone Two, you will further explore how the solderless breadboard works. Additional information on solderless breadboards can be found in the Module Two readings and resources.

Step 1: Prepare the Solderless Breadboard

Components

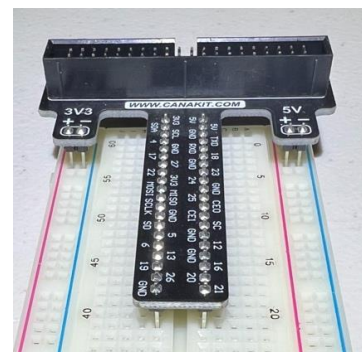
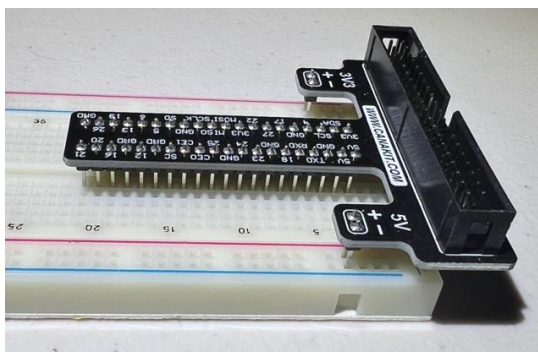
- Solderless breadboard
- GPIO breakout board

Remove the GPIO breakout board from the box it came in (it shares a box with the ribbon cable already installed on the Raspberry Pi).



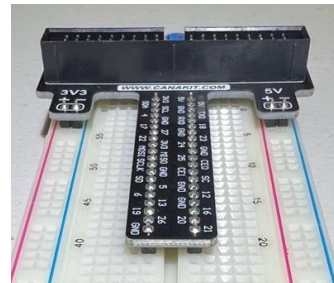
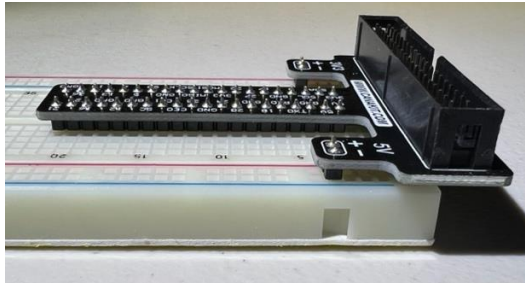
Carefully remove the GPIO breakout board from the foam backing it is attached to and line the pins up on the solderless breadboard. The pins beneath the 3v3 arm should align with the pins on Row 4 of the breadboard's left-hand power bus. The pins beneath the 5V arm should align with the pins on Row 4 of the breadboard's right-hand power bus. The pins running down the length of the GPIO breakout board should fall on either side of the central divide from Row 2 through Row 21.

Note: The numbers on the right-hand side of the breadboard denote the row numbers. These numbers start with 0, not 1.



Once you have the pins aligned, carefully press the GPIO breakout board into the solderless breadboard.

Note: Do not force the board. If it does not move with steady pressure, check to ensure that none of the pins are bent or misaligned. Once pressed into the breadboard, you should no longer be able to see the header pins on the bottom of the GPIO breakout board.



Step 2: Assembling Your First Circuit

Components

- 2 x male-to-male jumper wires
- 1 x 220 ohm resistor (stripe pattern: red, red, brown)
- 1 x red LED light

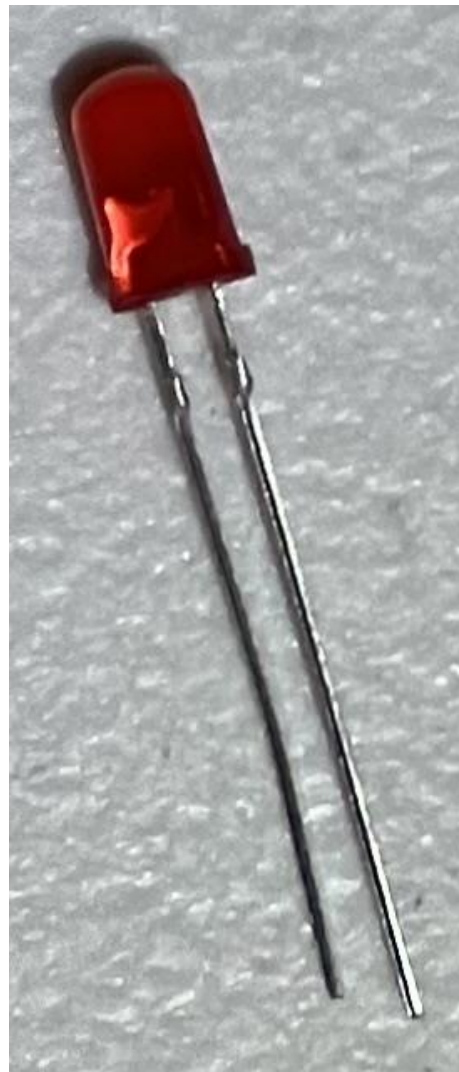
Note: You may have to gently pull the male-to-male jumper wire to separate it from the bundle of jumper wires. The kit comes with a color-coding chart to help you identify the value of the resistors included in your kit.



Male-to-male
jumper wire



220 ohm
resistor



Red LED light

Of the three components you are using in this step, the red LED is specific about how it is used in the circuit. The longer leg of the LED is called the anode (+) and needs to be connected to the GPIO pin, where it will receive voltage when the GPIO pin is set to high (sometimes referred to as True) in Output mode. The shorter leg of the LED is called the cathode (-), which will be connected to the resistor that will be tied to the ground pin on the GPIO header.

You will use pins 3 (ground) and 6 (GPIO 18) on the right side of the GPIO header. Choose one jumper wire and connect Column A, Row 4 (your ground pin) to Column E, Row 30.

Note: The columns in this bank of pins were labeled A, B, C, D, and E from right to left at the top of the board.

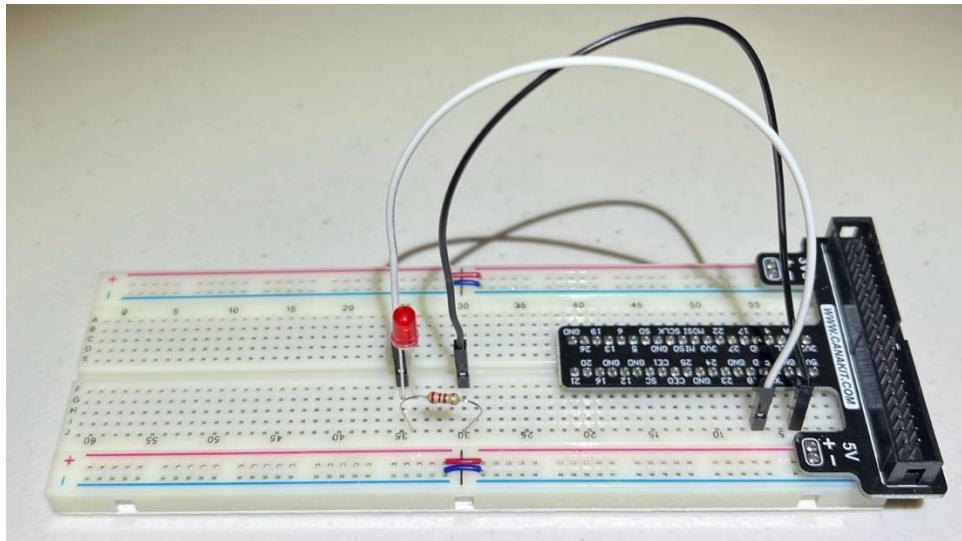
Choose one jumper wire and connect Column A, Row 7 (this should be the sixth pin on the right-hand side, labeled 18 on the breakout board) and connect it to Column E, Row 35.

Take the LED and connect the anode (long leg) to Column D, Row 35, and the cathode (short leg) to Column D, Row 34.

Take the resistor, gently bend the wire legs on each side, and insert one side into Column A, Row 34, and the other in Column A, Row 30.

Note: Do not let the leg of the resistor physically contact the LED, as this could cause a short and damage your circuit or your Raspberry Pi.

At this point, your board should look like this:



Step 3: Power Off Your Raspberry Pi

You must shut down and unplug your Raspberry Pi to continue. To safely shut down the operating system running on your Raspberry Pi, enter the following command from a terminal or the console of the Raspberry Pi:

```
sudo halt
```

You will have to enter your password as you are executing the halt command with root privileges by calling it with *sudo*. Wait at least two minutes after your terminal window closes before unplugging your Raspberry Pi.

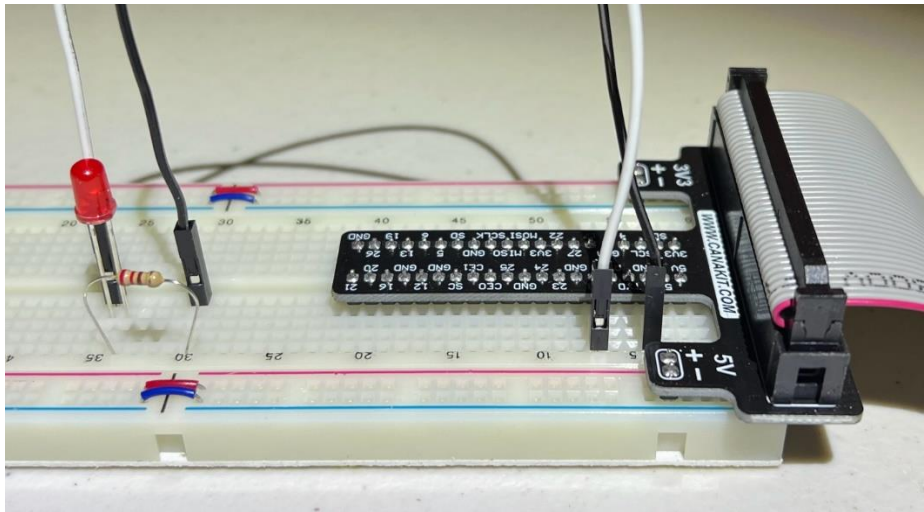
Stop: Raspberry Pi must be unplugged before proceeding. Never connect the GPIO cable while the Raspberry Pi is connected to power.

Step 4: Connect the GPIO Ribbon Cable to the GPIO Breakout Board

Components

- Raspberry Pi with installed ribbon cable
- Solderless breadboard
- GPIO breakout board

Carefully line up the connector on the end of the ribbon cable with the socket on the GPIO breakout board. Due to the notch in the connector, it will only fit in one orientation. While supporting the GPIO breakout board underneath the connector, press the connector into the socket until it is fully seated.



Step 5: Power Up Your Raspberry Pi

Plug in your Raspberry Pi. When it has completely booted, log in to the Raspberry Pi.

Step 6: Testing Your Circuit

There are three script templates that are part of the source code for Module One that you will be using to test your circuit in this lab. These templates are SimpleBlink.py, GPIO-18-ON.py, and GPIO-18-OFF.py. If you have not already transferred the source code ZIP file to your Raspberry Pi, you can find instructions on how to do so in the Raspberry Pi Setup Tutorial.

Once transferred to your Raspberry Pi, each script can be run in the following manner:

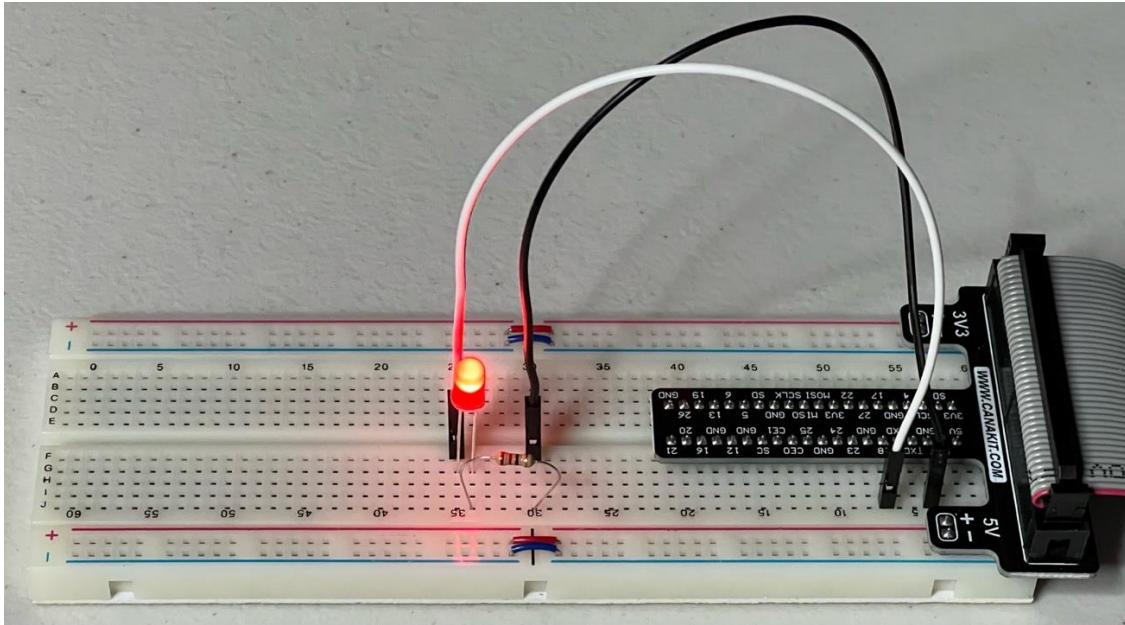
```
sudo python3 GPIO-18-ON.py
```

You must run the scripts with *sudo*, as they require access to a device (/dev/mem) with restricted access. The functionality of the scripts is straightforward.

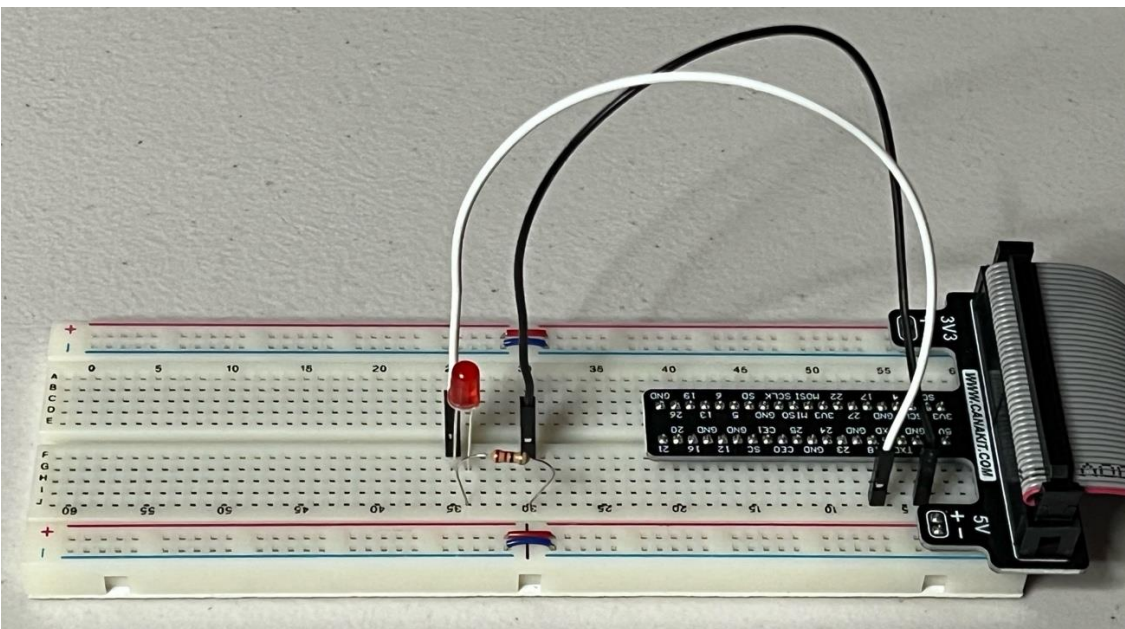
- GPIO-18-ON.py enables an output signal on GPIO Line 18.
- GPIO-18-OFF.py disables the output signal on GPIO line 18.

- SimpleBlink.py combines the functionality of the two previous scripts in a loop to make the LED blink.

Each of the scripts is fully documented for your reference. Once you have your scripts available on your Raspberry Pi, you should be able to execute the GPIO-18-ON.py script, and the result should look like this:



Running the corresponding GPIO-18-OFF.py script should return the LED to the off state:



The SimpleBlink.py script is provided as part of the source code for Module One to give you an idea of the next steps with this type of functionality. Please read the script and the comments within the code before running it as a test.

If your tests were successful, you now have a fully configured test environment. This will allow you to complete Milestone One in Module Two.

Troubleshooting

If your LED does not light up when you run the GPIO-18-ON.py script, shut down and unplug your Raspberry Pi before performing the following verifications:

1. Make sure that your jumper wires are connected properly. Ensure that the jumper wire connecting GPIO-18 to the LED is plugged into the right connection point and connected to the longer leg of the LED.
2. Ensure that both legs of your LED are seated on the solderless breadboard.
3. Verify that your LED is installed correctly with the anode (longer leg) connected to the jumper wire and the cathode (shorter leg) connected to the resistor.
4. Verify that your second jumper wire connecting the resistor to a ground pin is plugged in and connected properly on both ends.
5. Try using a different LED. Your kit has extras.
6. Try replacing the jumper wires one at a time. Your kit has extras.

If all these items are correctly configured and you're still having difficulties, take one or more pictures that indicate your connections and send them to your instructor for further assistance.